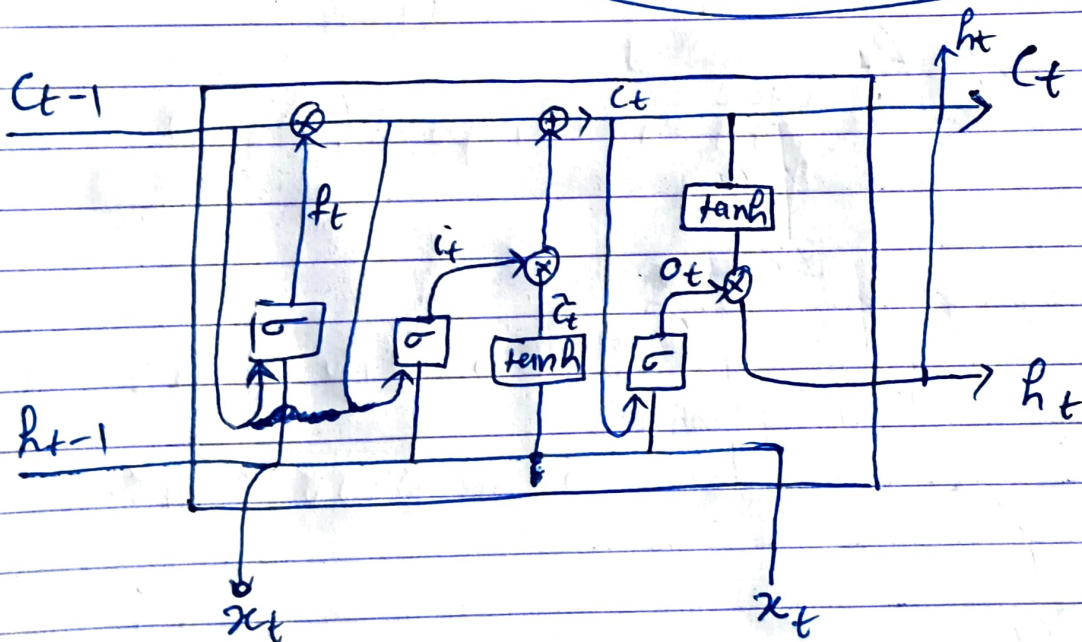


Training Data with LSTM RNN

variants of LSTM

Peephole connections:- we let the gate layers look at the cell state.

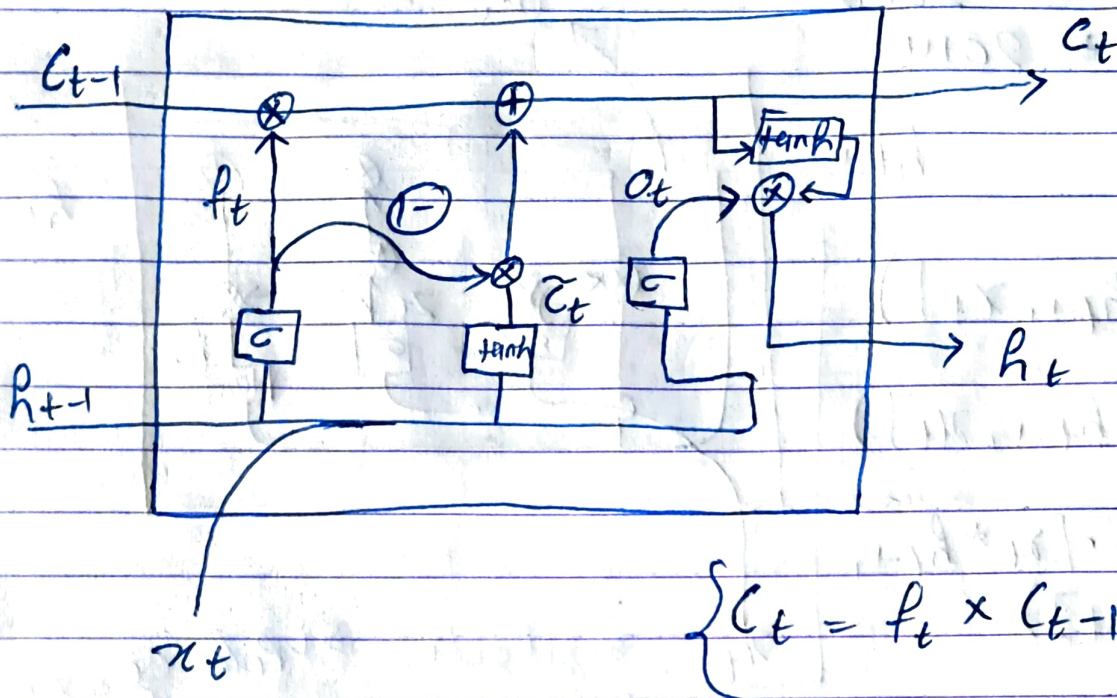
forget gate
input gate
output gate } $\Rightarrow$ Peephole connection.
from memory cell to



$$\left. \begin{aligned} f_t &= \sigma(w_f[C_{t-1}, h_{t-1}, x_t] + b_f) \\ i_t &= \sigma(w_i[C_{t-1}, h_{t-1}, x_t] + b_i) \\ o_t &= \sigma(w_o[C_{t-1}, h_{t-1}, x_t] + b_o) \end{aligned} \right\} \text{equation.}$$

Another variation:-

→ Coupling forget gate and I/P Gates.



instead of separately deciding what to forget and what ~~to~~ we should add new Info, we make this decision together.

goal:- we only forget when we're going to I/P something in its place.

⇓

we only I/P new values to the state when we forget something older.

GRU $\Rightarrow$ Gated Recurrent Unit

LSTM $\rightarrow$ 1980

variants $\rightarrow$ 2000

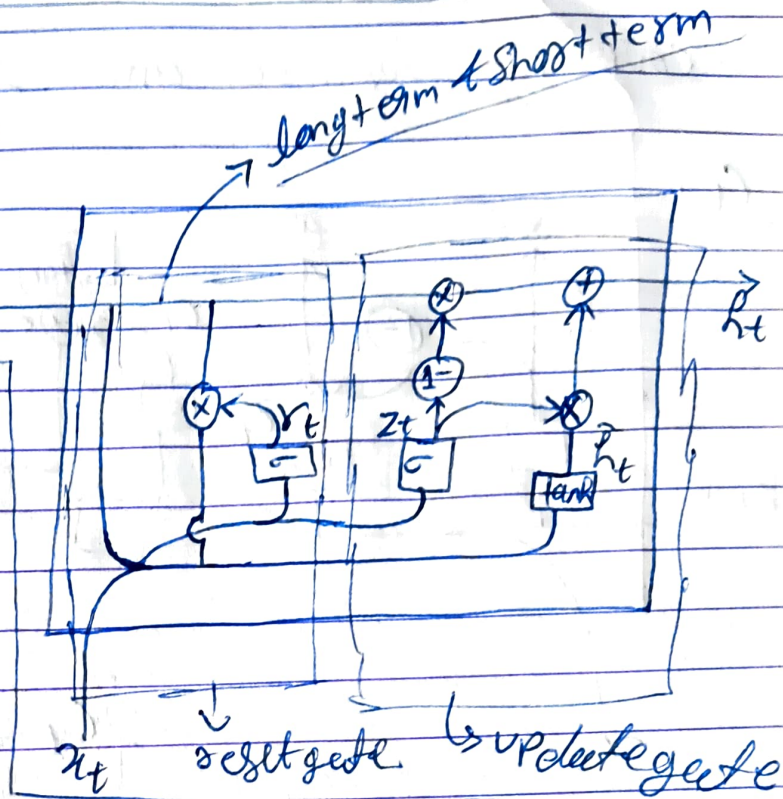
GRU $\rightarrow$ 2014

$$z_t = \sigma(w_{z0} \cdot [h_{t-1}, x_t])$$

$$r_t = \sigma(w_{r0} \cdot [h_{t-1}, x_t])$$

$$\tilde{h}_t = \tanh(w \cdot [r_t * h_{t-1}, x_t])$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

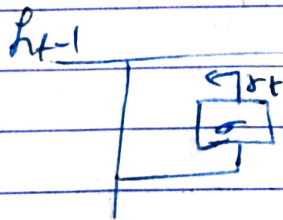


$z_t \rightarrow$ update gate.

$r_t \rightarrow$ reset gate.

$\tilde{h}_t \rightarrow$ temporary hidden state.

Reset gate (r_t)



$\rightarrow$ Resetting some info from $h_{t-1} \Rightarrow$ memory

$\downarrow$
LSTM + SIM

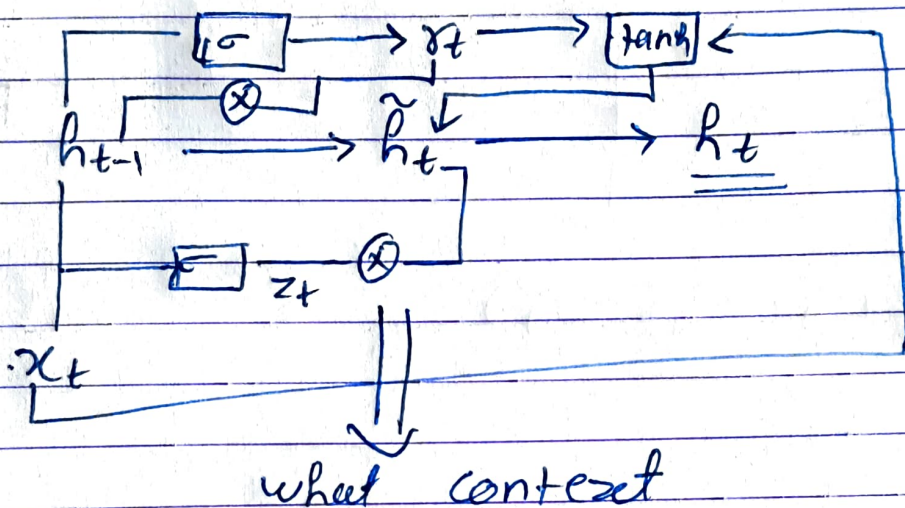
$$h_{t-1} \Rightarrow [0.6 \quad 0.5 \quad 0.3 \quad 0.9]$$

⊗

$$x_t \Rightarrow [0.2 \quad 0.4 \quad 0.8 \quad 0.2] \rightarrow \text{resetting these \% of values.}$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$[0.12 \quad 0.20 \quad 0.24 \quad 0.18]$$



what context

in the transition of h_{t-1} to h_t

